Learning from Comparison Data

M.Tech Project: Phase I

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Choice Sets and IIA

Travel from IIT Madras to the airport

Original choice set
Car, taxi, bike, bus

Expanded choice set
Car, taxi, bike, bus, walking

Independence of Irrelevant Alternatives (IIA)

Under IIA, the relative odds of choosing car over taxi stay the same when walking is added. A context effect occurs when the available alternatives change those relative odds.

Seshadri et al. (2019), *Discovering Context Effects from Raw Choice Data*.

Bradley–Terry Estimation

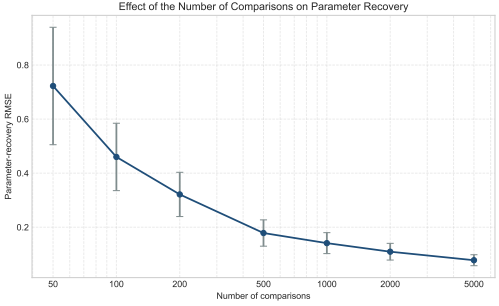
For items i and j, the probability that i defeats j is

$$P(i \succ j) = \frac{\exp(\theta_i)}{\exp(\theta_i) + \exp(\theta_j)}.$$

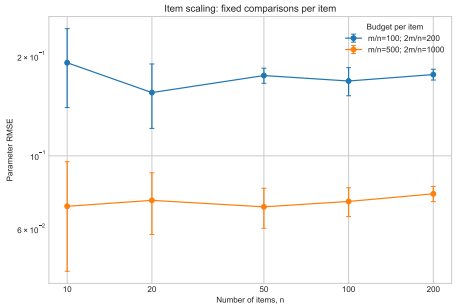
- Fit the scores by minimising the average negative log-likelihood.
- Centre the scores so their sum is zero.
- A finite MLE exists when the directed win graph is strongly connected.

Bradley–Terry Parameter Recovery

More comparisons with 10 items



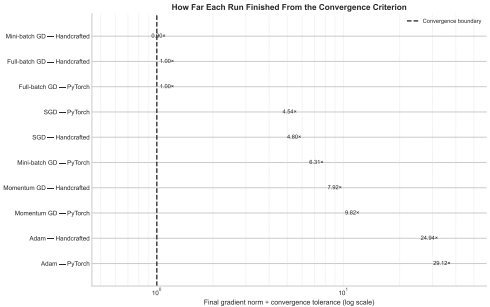
More items at fixed m/n



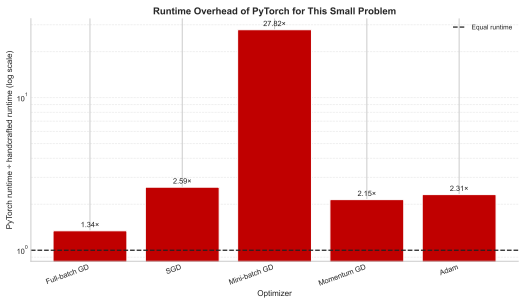
RMSE falls as comparisons increase. At fixed m/n, it stays roughly stable as the number of items grows.

Comparison of Optimisation Methods

Distance from convergence



PyTorch / handcrafted runtime



Recovery errors were similar, but convergence and runtime differed across methods.

Context-Dependent Model

Utility of item x in choice set C and its choice probability:

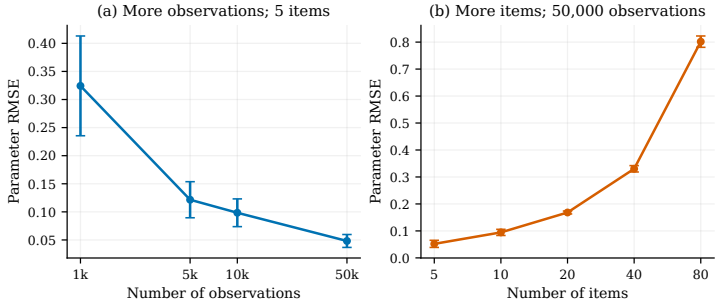
$$u(x \mid C) = \sum_{z \in C \setminus \{x\}} u_{xz}, \qquad P(x \mid C) = \frac{\exp(u(x \mid C))}{\sum_{y \in C} \exp(u(y \mid C))}.$$

Full CDM
Estimate a parameter for each ordered pair. The centred model has $n(n-1)-1$ free parameters.

Low-rank CDM
Write $u_{xz} = t_x^\top c_z$ for vectors in $\mathbb{R}^r$. The implementation stores $2nr$ trainable entries.

Full CDM Parameter Recovery

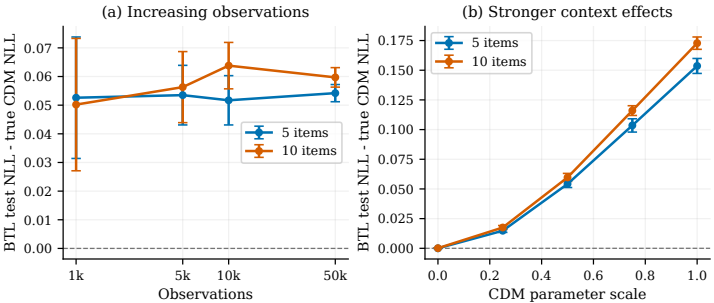
Synthetic choices from pairs and triples. The model is fitted on 80% of the observations.



More observations improve recovery for five items. With 50,000 observations fixed, recovery worsens as the number of items increases.

BTL/Luce Fitted to CDM-Generated Choices

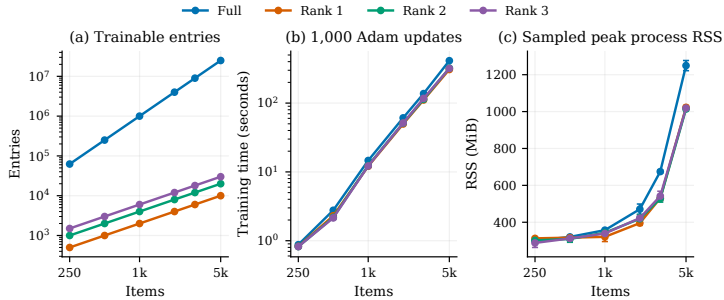
Prediction gap = fitted BTL test NLL − generating CDM test NLL.



More observations do not remove the gap in these experiments. The gap increases with the CDM parameter scale.

Computational Scaling of Full and Low-Rank CDM

250–5,000 items. Each fit uses 1,000 observations and 1,000 Adam updates on the CPU.

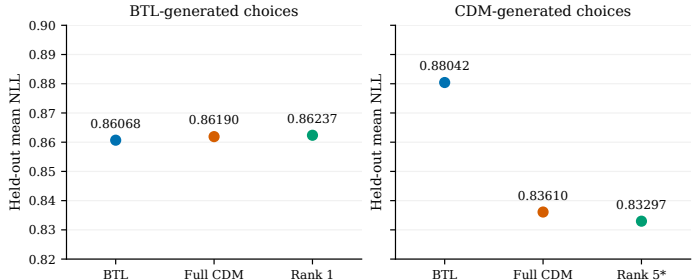


RSS is resident set size, the RAM used by the running process.

Low-rank CDM stores fewer parameters, but fitting time remains similar in this implementation.

Unified Comparison on Held-Out Choices

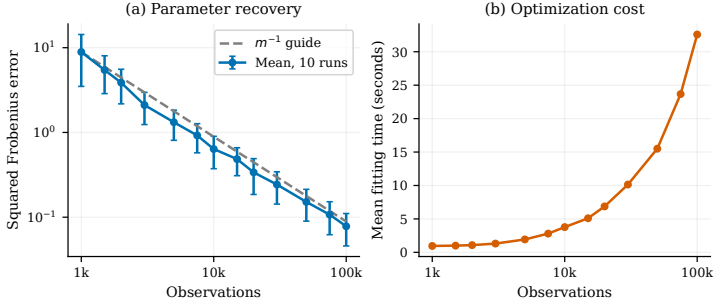
10 items, 10,000 observations per generator, 70/15/15 split. Lower test NLL is better.



The models perform similarly on BTL-generated choices. Both CDM fits have lower test NLL on CDM-generated choices, although the selected rank-five fit has not converged.

Six-Item CDM Parameter Recovery

Choice sets contain two to six items. Results average ten runs at each observation count.



The sum of squared parameter errors decreases as observations increase, while fitting time rises.

Following Seshadri et al. (2019).

Conclusion

Parameter recovery
More observations improved recovery. Increasing the number of items under a fixed data budget made estimation more difficult.

Context effects
Fitting BTL/Luce to the tested CDM-generated choices left a predictive gap.

Low-rank CDM
Fewer stored parameters did not translate into the same reduction in runtime or total process memory.

Further work
Check the comparisons across independent datasets and improve optimisation.